

Method Pilot: Test the Test

BUILD • Science • 40 minutes

I can carry out or direct a small method pilot, identify a procedural problem and revise the plan ready for W14.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

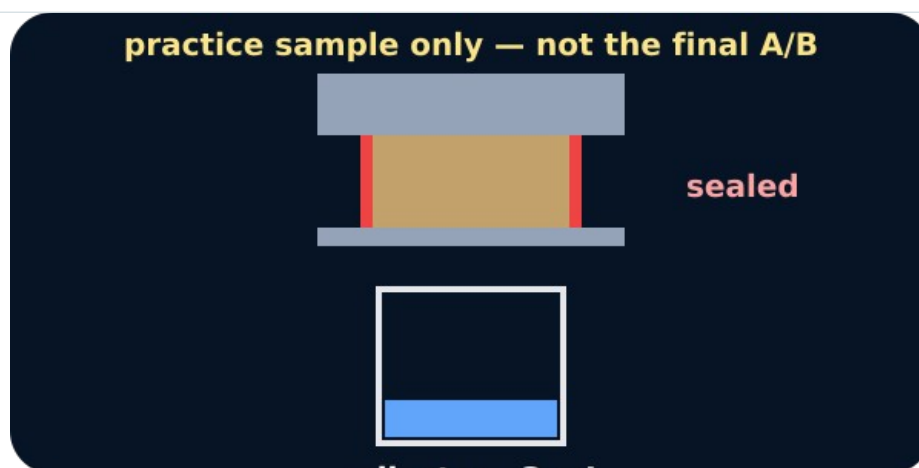
Word	Meaning
pilot	small practice run
fault	a problem in the method
revise	change the plan clearly

First idea

Pilot P1 uses one practice sample. One millilitre leaks around the collar edge. What should happen next?

- A. Use the tray water to decide the practice rock is highly permeable.
- B. Record the edge leak as a method fault, revise the seal check and pilot the repaired method again or use the fixed route.
- C. Hide the leak and copy the expected value so the method looks ready.

My first idea and reason:



A practice run on the bench

We do • choose, explain, check

Which pilot response improves the future investigation without creating a false conclusion?

A. A leak occurs, so record 'practice rock is permeable' and start W14.

B. A leak occurs, so add 'press and inspect the seal before adding 5 mL'; rerun the practice route and keep final A/B results blank.

C. A leak occurs, so change the sample, water volume and time together until a neat value appears.

My choice: _____ Evidence or reason:

Pilot fault-and-revision desk

Route each pilot note to KEEP, REPAIR or W14 READINESS. Do not route any card to a final rock conclusion.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
KEEP IN METHOD V2	
REPAIR THE METHOD	
W14 READINESS DECISION	

Activity cards

1 • Practice label	2 • Volume step
The pilot piece is clearly marked PRACTICE P and stored separately from final Samples A and B.	The method states add exactly 5 mL using the marked needle-free tool.
3 • Stop rule	4 • Pilot fault P1
Stop for a leak, unstable rig, sample damage, flaking, dust or unclear measurement and switch route.	One millilitre runs around the collar edge into the tray before the timer ends.

5 • Table fault

The collector column is headed Amount but gives no unit.

6 • Timing fault

The instruction says wait a while and two pupils choose different end points.

7 • Ready decision

Practice P2 holds its seal, follows 5 mL and 120 seconds, records mL clearly and triggers no safety stop.

8 • Repair then review

The second practice still leaks or gives an unreadable collector value.

9 • Fixed-route decision

A stable, measurable physical setup is unavailable, so the approved W14 route uses the supplied repeated data and the same reasoning table.

State: The pilot found ___. We revised ___. The W14 route is ready / needs review because ___. The pilot cannot tell us ___.

Your lesson record

Carry out or analyse one practice pilot, record a method fault or successful check, revise the method and issue a W14 readiness decision without a final rock conclusion.

Model helps us see

Use: The pilot showed ___. This affects the method because ___. We revised ___ to ___. For W14, the route is ___.

Model does not show

A successful pilot only shows that the procedure can be followed and recorded under practice conditions. It does not prove the final samples will produce a pattern or answer the investigation question.

Supported

Sequence the practice steps, choose one fault card and one matching repair, then select ready, review or fixed route for W14.

Use: The pilot showed ___. We changed ___. For W14 we will ___.

Standard

Complete the pilot log, justify one precise revision and issue an evidence-based readiness decision.

Keep method observation, revision and readiness in separate boxes; leave final conclusion blank.

Stretch

Evaluate whether the revision fixes the identified fault without creating a new variable, then specify the W14 approval conditions.

Explain why a successful pilot increases procedural confidence but not confidence in a final sample pattern.

Evidence target

Pilot checklist, one procedure observation or fault, method V1 and visibly revised V2, safety/access decision, W14 readiness card and a statement of what the pilot cannot conclude.

Exit, Audience and evidence • W13B

Supported: Complete or point: The pilot found ___. We changed ___. W14 route ___.

Standard: Describe the method fault, revision and readiness decision without giving a rock conclusion.

Stretch: Explain why the pilot can approve a procedure but cannot support the final A/B permeability claim.

What I said, and what it changed

Next I will _____.

Adult witness note

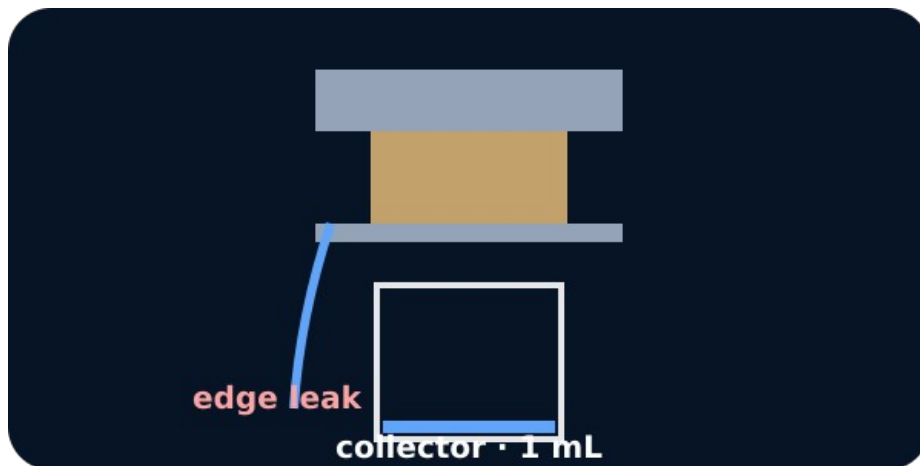
What was observed, and with what level of independence?

My evidence and explanation

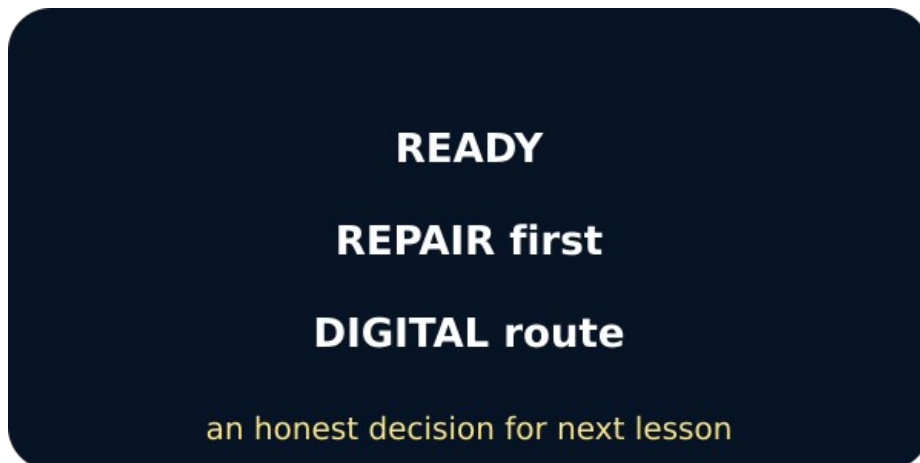
What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



An edge leak during the pilot



Three readiness decisions

Exit • one next step

After the adult has listened, what will you keep, improve or check next?
